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SHORT COMMUNICATION

Investigation on extremely dilute solution of PEO by PFG-NMR

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Abstract The technique of pulsed-field gradient nuclear magnetic resonance (PFG-NMR) was applied to study the solution properties of a series of low molecular weight poly (ethylene oxide). The self-diffusion coefficients of solutions from semi-concentrated to extremely diluted were measured, leading to a critical concentration. When the concentration of solution is higher than the value of critical concentration, the diffusion coefficient of the solute decreases as the concentration increases and remains the same when the concentration is lower than it. This critical concentration agrees well with the definition of dynamic contact concentration (C_s) and confirms indirectly the Flory's scaling law between the molecular weight and D_0. In addition, the influences of molecular weight and terminal groups on C_s were discussed. All the diffusion coefficients determined at extremely dilute condition were equivalent to the diffusion coefficients at infinite concentration (D_0), from which the polymer coil size was estimated.

Keywords Poly(ethylene oxide) · PFG-NMR · Self-diffusion coefficient · Dynamic contact concentration

Introduction

Understanding the physical properties of polymer solution has been a major topic and is of great significance in physics, chemistry, chemical engineering, and material science [,].

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The self-diffusion coefficient (D), which provides a direct measure of the translational motion of polymer chains, is vital in characterizing the dynamics of polymer solution [–]. The measurement of D has got a number of applications, such as evaluating molecular weight, estimating the distribution of molecular sizes, studying the inter-molecular interaction in solution, and so on [,].

The scaling of diffusion coefficients as molecular weight increases has received considerable attention. Flory's scaling law [ ] for polymer diffusion gives the relationship of self-diffusion coefficient at infinite dilution and the polymer molecular weight. It has been proved remarkably successful. Unfortunately, up to now, there is still not a method that can directly measure the self-diffusion coefficient at infinite dilution. In the dilute regime, it is a common practice to linearly extrapolate the measured self-diffusion coefficients towards zero concentration. The method needs an assumption that the dependence of self-diffusion coefficient on concentration in extremely dilute regime should be the same as in the dilute regime. Through a carefully designed dynamic light scatting experiment, Wu [ ] demonstrated that the concentration dependence in extremely dilute regime is different from that in diluted regime, indicating that the linearly extrapolate is not adequate. Thus, further study on the solution properties of polymers in extremely dilute regime is necessary.

Pulsed-field gradient nuclear magnetic resonance (PFG-NMR) method [ ] was developed by Stejskal and Tanner. By using a pair of pulsed-field gradients, one for dephasing and the other for refocusing the magnetizations, the self-diffusion coefficient can be calculated through the decay of the observed signal intensity. Though a number of versions of this method have evolved to be used for quite different purposes, the concept and theory nevertheless remain the same [ ]. The PFG-NMR method may be used to measure the self-diffusion coefficient of molecules by detecting the decay rate of the magnetization as a function of PFG strength. This method is in general not limited by the size of molecules. For samples of low molecular weight and long relaxation time, the measurement is usually sensitive and convenient. Besides, PFG-NMR can measure all the molecules in the studied system with the same experiments. It is possible to determine the width of molecular weight distribution in a multi-disperse polymer system [ ]. On the other hand, because the self-diffusion coefficient is calculated from NMR signal intensities, the influence of the impurities can be reduced greatly.

In this work, a series of poly(ethylene oxide) (PEO) with low molecular weight was selected for studying solution properties. Linear PEO is an unusual polymer in that it is soluble in various solvents. Possible aggregation of the polymer in solution makes it possible to determine the structures and dynamics of additional interests. A considerable amount of data is available on diffusion of PEO [,]. Much of the previous work has focused on the dilute and semi-dilute solution. For example, Waggoner et al. [ ] studied the self-diffusion of PEO with different molecular weights and found that even under very low molecular weight, PEO dilute solution obeys the Flory's scaling law. Alami et al. [ ] studied the aggregation behavior of PEO with hydrophobic terminal groups in dilute solution. The properties in extremely dilute solution are of great challenge. In this paper, the PFG-NMR method was used to study the solution properties of low molecular weight PEO in dilute to extremely dilute region through the measurement of self-diffusion coefficients.

Experimental

Materials

Hydroxyl groups ended PEO samples with molecular weight of 1,500, 2,000, 4,000, and 6,000 and methyl groups ended PEO sample with molecular weight of 2,000 were purchased from Fluka. The extremely dilute solutions for NMR experiments are prepared by following steps: a series of standard $CHCl_3$ solutions were prepared, a certain amount of solution was extracted to NMR tubes using pipette, and then these NMR tubes were placed in vacuum at room temperature until the solvent evaporated. After that, 0.5 ml D_2O was added as new solvent. The samples were oscillated for 30 min and then hold still for 12 h before NMR measurement.

NMR methods

NMR measurements were performed on a Bruker-DSX 500 MHz spectrometer equipped with a probe of maximum PFG=53.5 G cm^{-1} in the *z*-direction. Correspondingly, the maximum effective PFG for a sine shape is 34.1 G cm^{-1}. Heavy water was used for locking the magnetic field and the temperature was set at 298 K (±0.1 K). To determine the diffusion coefficients reliably, every NMR experiment was repeated three times.

Diffusion coefficients were determined using a BPP-STE [ ] pulse sequence $[(\pi/2)–\tau_2–(\pi)–\tau_2–(\pi/2)–\tau_1–(\pi/2)–\tau_2–(\pi)–\tau_2$–acquisition] with sine-shaped bipolar gradient pulses of 1 ms duration. The pulse sequence is shown in Fig. . Here, the sine-shaped PFG is used to reduce the effects of eddy current while the bipolar PFGs with a 180° pulse in the middle are applied to alleviate the effects of RF inhomogeneous. The magnetization from the molecules near the edges of the RF coil, which experience a flip-angle much smaller than 180°, will be dephased before detection. The interval between the gradient pulses (Δ) was set to 100 ms. For each diffusion time, 16 different values of gradient strength were used varying from 2% to 95% of the maximal gradient strength (35.1 G/cm). At every gradient strength, 32 scans were acquired with a recycle delay of 5 s. Spectra were processed with a Lorenzine weighting function of 0.3 Hz line broadening. The gradient strengths were calibrated using a D_2O/H_2O (1:1) sample at 298 K. If diffusion occurs during the time of Δ, incomplete refocusing of the magnetization arises. It causes an additional decay of the signal intensity *I*, which is given

$$\ln(I/I_0) = -D\gamma^2 G_e^2 \delta^2 (\Delta - \delta/3 - \tau/2) \quad (1)$$

where $G_e = (1/\pi)\int_0^\pi \sin(x)dx = (2/\pi)G_{max}$ is the effective PFG strength of a sine shape, *D* is the self-diffusion coefficient, γ is the magnetogyric ratio of the nucleus, δ is

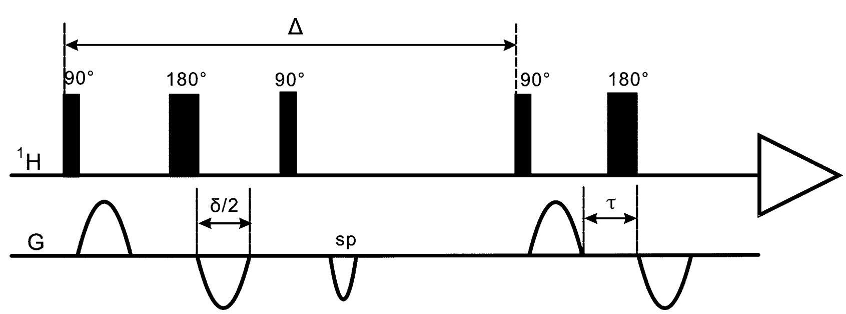

Fig. 1 The pulse sequence of BPP-STE for measuring diffusion coefficient. *sp* spoil pulse

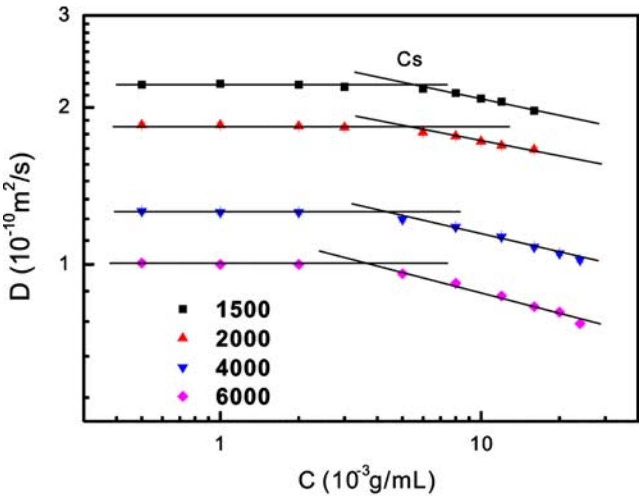

Fig. 2 Diffusion coefficient of PEO versus the concentration, where a critical concentration is defined at the crossing of extremely dilute and dilute region. In the extreme dilute region, a horizontal line with the average value of the data points is assumed and in the dilute region, least square fitting is used with the root mean square (*RMS*) error (Table)

the width of the gradient pulse, and Δ is the time from the start of the first gradient pulse to the start of the third pulse. In Eq. , a delay τ (100 μs) is introduced for gradient recovery, which is neglected in the calculation due to its small value compared with Δ=100 ms. A plot of $\ln(I/I_0)$ verse G_e^2 gives a straight line with a slope proportional to D.

Results and discussion

The self-diffusion coefficient (D) of PEO with different molecular weight was measured. Figure shows the dependence of D on concentration. It is apparent that the variation of D with the decreasing of concentration can be divided into two regions. In the relative higher concentration region, D increases as the concentration decreases, while at very low concentration region, D nearly remains the same when the concentration further decreases. This phenomenon shows that the concentration dependence of D is different in different concentration regions. This leads to a critical concentration at the crossing of two straight lines (Fig.). Therefore, the PEO concentration region has been divided into two regions according to the concentration dependence of self-diffusion coefficient. The self-diffusion coefficient changes with concentration linearly in one region and keeps constant in the other.

For classifying the concentration region of polymer solutions, de Gennes [ ] put forward a critical overlap concentration C^* as the turning point from dilute to concentrated solution. It means that from this concentration, polymer coils begin to overlap physically. This is widely accepted. Until the 1980s, Qian et al. [ ] proposed a different critical concentration—dynamic contacting concentration (C_s) from fluorescence spectra. This critical concentration characterizes a point at which a polymer molecule would begin to feel the existence of the other polymer molecules when the concentration is higher than C_s. The C_s was defined as the boundary point between dilute and extremely dilute solution. They thought that when the concentration was higher than C_s, the collision probability was high enough and it could not be omitted. As a result, the solutes in the solution were no longer single chains. Cheng et al. [ ] put forward a cluster theory, the main idea is that the solutes can form clusters automatically because of the random collision and the clusters may affect the solution properties greatly. When the clusters increase to a certain degree, the solution properties begin to change. The dynamic contact concentration was then indicated by SEC [ ] and dynamic light scattering [ ] experiments at linear polymer systems. We have also observed C_s at hyperbranched polymer systems by viscosity method [ ].

In this work, the physical picture of our results coincides with the dynamic contacting concentration. The observed critical concentration, we think, should be C_s. According to this theory, the phenomenon we observed can be explained as follows: on the extremely dilute region, the PEO molecules are single chains, and one chain may not be affected by the other chains in the time scale of diffusion experiment (~100 ms). Therefore, the friction among solute molecules can be omitted and the friction of solution mainly comes from

Table 1 Diffusion coefficients at extremely dilute concentration (D_0) of PEO derived by two different methods, where least square fitting is used with a root mean square (RMS) error

Molecular weight	D_0 (10^{-10} m^2/s)				
	Calculated by extrapolate	RMS error	Calculated by our method	RMS error	Error of different method (%)
1,500	2.44	0.009	2.21	0.013	10.56
2,000	1.89	0.004	1.84	0.022	2.77
4,000	1.44	0.009	1.26	0.006	14.09
6,000	1.16	0.008	1.01	0.012	15.29

Table 2 Physical properties (D_0, C_s, C^*, and R_H) of PEO measured by PFG-NMR

Molecular weight	D_0 (10^{-10} m^2/s)	C_s (10^{-3} g/mL)	C^* (g/mL)	R_H(Å)		
				Calculated by extrapolate	Calculated by our method	Error (%)
1,500	2.21	4.49	0.11	8.94	9.89	9.61
2,000	1.84	4.08	0.09	11.52	11.84	5.56
4,000	1.26	3.92	0.05	15.14	17.27	12.33
6,000	1.01	3.63	0.04	18.79	21.66	13.25

the friction between solute molecules and solvent molecules. According to the Stokes–Einstein equation [ ]:

$$D = \frac{kT}{f} \tag{2}$$

where K is Boltzmann constant and T is absolute temperature; the friction coefficient f can be determined

$$f = 6\pi\eta R_H \tag{3}$$

where R_H is the hydrodynamic diameter of solute molecule, η is the solution viscosity.

To obtain the self-diffusion coefficient at infinitesimal concentration (D_0), people usually extrapolate the self-diffusion coefficient from dilute solution to zero concentration. While from our results, the extrapolating is not exact. The self-diffusion coefficient keeps a constant at extremely dilute region. We think that this constant should be the truly D_0. The observed D_0 was listed in Table . The difference between D_0 obtained from extrapolating and the truth D_0 may vary depending on molecular weight. For polymer with higher molecular weight, the error may be up to 8% [ ], while for polymer with smaller molecular weight (as our system) the error may be up to *15%*.

The C_s can be determined by experimental method, as shown in Fig. . The horizontal line shows the trend of self-diffusion coefficient at extremely dilute region and the other line shows the trend of self-diffusion coefficient at dilute region. At the point of intersection C_s is determined. The data are shown in Table .

Normally, the friction between solute and solvent is affected by the shape and size of solute molecules, as Stokes Eq. shows.

For polymers of high molecular weight, the end groups hardly influence the solution properties. While for polymers of low molecular weight, the solution properties may be affected by the end groups because their portion in the molecule becomes higher. In order to study if the end group would affect the solution properties in our system, we selected PEO with the same molecular weight but different end groups, i.e., hydrophobic methyl and hydrophilic hydroxyl group as our samples, and measured the diffusion coefficients of both samples. As shown in Fig. , we can see that there is almost no difference in diffusion coefficient between the two samples, which means although the hydroxyl group may form hydrogen bond with the segments, the effect is rather

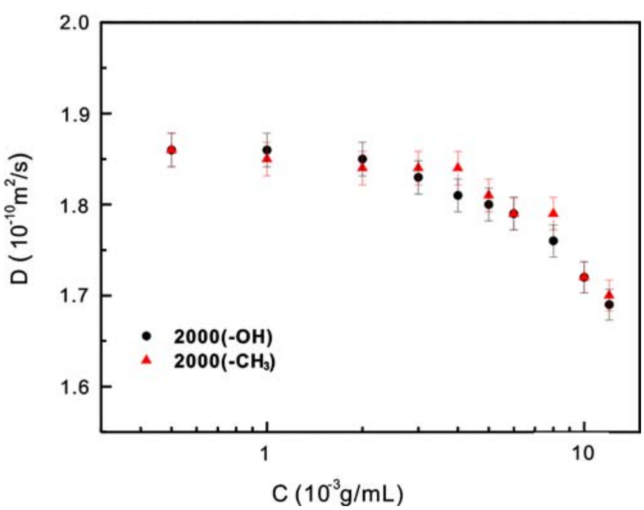

Fig. 3 Diffusion coefficient of PEO versus concentration, where *filled circle* and *filled triangle* denote PEO with end group hydroxyl and methyl, respectively

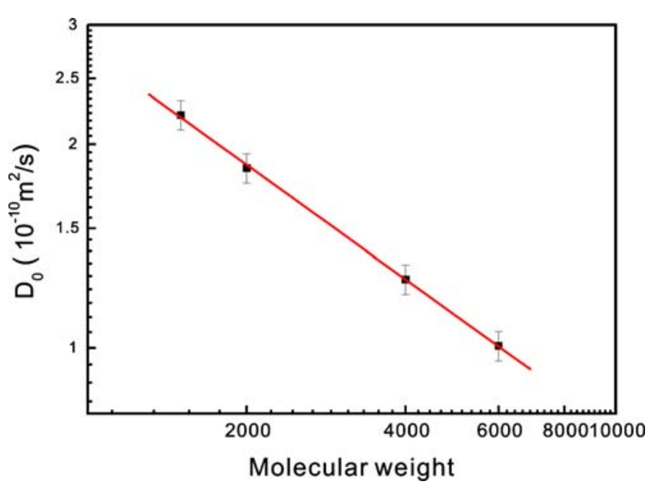

Fig. 4 Dependence of D_0 on the molecular weight for the sample of PEO

small. Comparing with the interaction of the segments, both the ability of hydroxyl groups to form hydrogen bond and the hydrophobic properties of methyl groups are not considerable in the concentration range of our study.

The relationship between D_0 and molecular weight was also studied. Flory's scaling law, derived from the hydrodynamic dependence of the diffusion coefficient is:

$$D_0 = KM^{-\alpha} \tag{4}$$

where K is a constant and the scaling coefficient α ranges from 1/2 for a θ solvent to 3/5 for a good solvent.

The effect of molecular weight on D_0 is shown in Fig. . From the data analysis, we find that the data agree well with the Flory's scaling law. The calculated index α is 0.56, which matches the properties of PEO quite well.

As we know, the polymer coil size can be determined from the self-diffusion coefficient. The Stokes–Einstein Eq. can be converted to:

$$R_{\mathrm{H}} = \frac{kT}{6\pi\eta D_0} \tag{5}$$

The hydrodynamic diameter of the PEO molecules in solution can be calculated according to Eq. . The calculated R_{H} was listed in Table . If D_0 was obtained from linearly extrapolate, its value is overestimated, which leads to an underestimated R_{H} For our samples, if the R_{H} was calculated from the D_0, which was obtained from linearly extrapolate, the error of R_{H} would be about 5～13%.

De Gennes [ ] pointed out a critical concentration as overlap concentration C^*, and the C^* can be calculated by the followed equation:

$$C^* = 3M/\left[4\pi N_{\mathrm{A}}\left(R_{\mathrm{g}}\right)^3\right] \tag{6}$$

Where N_{A} is the Avogadro's number and R_{g} is circular diameter. For flexible polymer as PEO, the relationship of R_{H} and R_{g} is $R_{\mathrm{H}}/R_{\mathrm{g}}$=0.665.

According to the obtained R_{H}, the C^* can be calculated. Table shows the C_{s} and C^* of PEO with different molecular weight. The C^* is one or two order of magnitude greater than the critical concentration we observed. This result further proves that the critical concentration we observed is not overlap concentration, but the dynamic contact concentration.

Conclusion

A critical concentration of PEO was observed with the method of PFG-NMR. When the concentration is lower than that, the self-diffusion coefficient remains the same. We believe that this critical concentration is the dynamic contact concentration, which corresponds to the turning point of dilute and extremely dilute solution. Thus, this is the first NMR measurement of the dynamic contact concentration. In addition, the D_0 obtained by our method is much more accurate than the one which was linearly extrapolated. The scaling behavior of diffusion coefficient was studied. The relationship between molecular weight and D_0 has been measured using the PEO with the molecular weight of 1,500, 2,000, 4,000, and 6,000. It obeys well with the Flory's scaling law described by $D_0 = KM^{-\alpha}$. With the Stokes–Einstein equation, the sizes of PEO molecular chain in an aqueous solution were also derived for the molecular weight of 1,500, 2,000, 4,000, and 6,000, respectively.

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